A Procedural Approach for Evaluating the Performance of Business Processes Based on a Model of Quantative and Qualitative Measurements

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Abstract. Evaluating the performance of processes is of vital importance if organizations are to seek continuous improvements. It is by measuring processes that data on their performance is provided, thus showing the evolution of the organization in terms of its strategic objectives. These results will serve as the basis for making better decisions, thereby leading to continuous improvement. The approach set out in this paper is prompted by the relative lack of empirical investigations into performance measures contained in the literature and the difficulties that organizations face when trying to verify the results of their business processes. Based on analyzing studies selected in a Systematic Review of the Literature, there it was found the need to propose a new approach to evaluating business processes that brings together elements and recommendations selected from the analyzed approaches. In the evaluation of the proposed approach, a case study is discussed, to verify its applicability.

Keywords: Strategic alignment · Business process management · Performance measurement

1 Introduction

Organizations undertake their activities in a context that is characterized as being highly competitive, complex and subject to rapid mutations in addition to which customers demand more and more and all of which arises from greater access to information. Business processes help organizations to see to it that their activities become more flexible and that they enhance the quality of their products and services, thereby making themselves suit the demands of the market and thus leading to the natural development of the Organization. Against this background, organizations need to integrated management by running an interactive system of processes that carry out work with a view to delivering value to their customers. From this perspective, the concept of Business Process Management (BPM) has emerged as a management approach that shifts the focus from functional units to controlling the performance of business processes in achieving their objectives.

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The results of the processes are directly linked to the mission and objectives of the organization, since the processes represent the implementation of the strategy. The focus of process management is underpinned by key business strategies that establish the direction of the organization. In this context, [2] reinforces that the need for measurement in process management is a critical factor. It uses measurements that make it possible to monitor the performance of processes, thus contributing towards monitoring processes, thus contributing to checking how well these processes meet the strategic objectives set.

According to [3], issues related to performance measures and to defining what should be measured in relation to business processes should be directly linked to the strategic priorities of each organization.

BPM initiatives need to be evaluated to check the alignment between the strategic, tactical and operational aspects of processes, thus making it possible to verify the results achieved in accordance with the objectives outlined. Business processes should be designed by the administration, after having established measures of performance (Powell et al., 2001), which should reflect the desired direction given in the strategic objectives, and serve as a basis for the control of processes. By so doing, this will make it easier to see if the objectives of the organization are being attained.

However, according to both [5, 6], the effects of BPM programs are often not easily visible as to whether or not value is being generated for organizations. This is because organizations are unaware of or do not have good control over the operation of their processes. The result of this is to create performance indicators that reflect departments' prompt results, defined by a functional management focused on a vertical view [7]. Thus, measures and evaluation of performance end up targeting the functional performance of departments and individuals when they should focus on the outcomes of the process.

According to [8], organizations have difficulty in defining what should be measured, and end up measuring less complicated aspects such as productivity, cost, time; and often neglect indicators linked to strategy. Based on their analysis of empirical studies, [7] verified there is a gap between the analysis, implementation and conduct of processes as well as between strategic management and operational running; for which they give as a possible explanation that a methodological orientation of systems that measure the performance of Business Processes is lacking.

[2] takes the view that the sustainability of a BPM initiative is adversely affected in several ways when the value added to the organization cannot be measured due to the fact that several BPM initiatives are not supported by reports or measures that managers who contribute to the decision-making process understand. In the analysis of [7], various models for assessing the performance of Business Processes use numerical parameters and artificial and simplifying measures in an attempt to assess these processes. However, various processes of a qualitative and non-deterministic nature end up not being evaluated effectively.

Arising from the idea that various processes are not evaluated or that this evaluation is difficult when using these measures, possible problems or performance results may remain invisible to managers, and thus make hardly any contribution to the decision-making process and consolidate the gap between strategy and business processes [7].

When searching for studies related to initiatives on measuring business process performance, the paper by [9] was evaluated. Based on a systematic review of the literature, this study classified measurement initiatives in business process management into two distinct categories: measures for models and measures for running or assessing BPM activities; the former being related to the static properties of the processes represented based on the models, and the latter, corresponding to measures related to running processes, thus allowing results obtained and expected to be compared with the strategic objectives. The study shows that 77% of performance measures are related exclusively to the models of the processes, thus giving evidence of the lack of models that aim to evaluate the results of the Business Processes themselves.

The measurements for running a BPM program are related to customer satisfaction, and have been studied in less depth in the Computer Sciences than in other areas. These measures seek to quantify how the process is carried out over time, thus allowing the results measured to be compared with the expected results and thus providing information that contributes to improving business processes [9], which are the focus of this research study.

What prompted this research is related to the difficulty for organizations of finding comprehensible measurements to assess whether the results of the business processes are reflecting the strategic objectives noting the paucity of empirical research on using models and performance measurement systems in the literature. This leads us to following research question:

Which approach best meets the evaluation of business processes, with a view to bringing about a greater alignment between process indicators and strategic objectives?

Based on the model proposed by [10], which was adapted from the GQM (Goal Question Metric) method, the overall objectives of this research are: (1) to analyze metric models or approaches for assessing business processes proposed in the literature; (2) to put forward an approach to evaluating business processes that contributes towards better aligning indicators of processes and strategic objectives; (3) to check its applicability from the standpoint of leaders, analysts and owners of the process; (4) to do all these in the context of business processes of a public organization.

The rest of the article discusses the systematic review of the literature in Sect. 2. Then, the Approach put forward in this paper is described (Sect. 3). Section 4 presents the results obtained by applying the Approach and in Sect. 5 final remarks are made, the contributions of the paper listed and suggestions for future studies made.

2 Systematic Review of the Literature

The Systematic Review methodology adopted in this paper is a form of research that uses the literature on a given subject as a source of data [11]. This approach is a means for identifying, evaluating and interpreting various research studies which are available and especially relevant to a specific research question, subject area, or phenomenon of interest.

According to [12], there are specific reasons that contribute to conducting a systematic review, such as:

- Consolidating evidence and results obtained in previous studies on a given topic;
- Identifying gaps in current research or theory, thereby offering a theoretical basis to improve research studies;
- Having the ability to provide background and theoretical models so as to position new themes and research opportunities appropriately;
- Enabling new hypotheses on a given research theme to be refuted, validated or developed.

The systematic review approach indicated by [12] is divided into three stages/ordered phases: planning, execution and analysis of the results.

The objective of this review is to analyze, more specifically, what the performance measures or indicators are and how they are being used in the context of managing business processes to ensure that processes and strategic objectives are aligned. Based on this analysis, we set out to answer the following Research Questions:

RQ1- What are the metrics and indicators that are being used when evaluating business processes?
RQ2- Do the studies have rules, guidelines or sets of guidance notes on how to use the metrics presented?
RQ3- What is the context in which the metrics are being used?

2.1 Planning the Review

In the automatic search strategy, a few key words taken from the research question were selected. The list of keywords was defined in conjunction with the supervisor of the research study and are given in Table 1 below.

Table 1. Key-words and synonyms.

Key-words	Synonyms
Metrics	Measured
Assessment	Evaluation
Measure	Measurement
Application	Utilization
Performance indicators	–

When formulating the search string, the keywords listed in the Table above were used. These describe the terms related to evaluating business processes according to [13]. The second group of words used was: “business process”, “BPM” and “business process management”. This group focuses a set of words related to business processes, which are the elements that will be measured.

The search string was formed by combining the terms described above with Boolean operators (AND/OR). The formulated string was entered on the search engines of digital libraries to obtain the studies related to the terms used.

Automatic searches were conducted in the following digital libraries:

- Association for Computing Machinery – ACM (http://portal.acm.org/portal/);
- IEEE Computer Society (http://www.computer.org/web/search);
- Emerald Insight (http://www.emeraldinsight.com/search/advanced);
- Science Direct (http://www.sciencedirect.com/science/search);
- Springer Link (http://link.springer.com/advanced-search).

The libraries of the IEEE Computer Society and the Association for Computing Machinery (ACM) are specific to the Computer area. Initially only the titles of the articles were read and, if in doubt, the summary was also evaluated.

The strategy of manual search was also used in order to complement the automatic strategy, thus allowing an investigation to be made in pre-defined locations that, possibly, could include articles related to the topic of interest.

When conducting manual searches, some conferences, journals and the references of the articles found were used. Only articles published after 2004 were evaluated. The search sites and the articles selected are listed below:

AMCIS (Americas Conference on Information Systems). Article selected: Year 2011: Limitations of Performance Measurement Systems based on Key Performance Indicators.

CAISE (Conference on Advanced Information Systems Engineering): No article was selected.

ICSE (International Conferences on Software Engineering): No article was selected.

BPM Conference (International Conference on Business Process Management): The following articles were selected:

Year 2005: Using software Quality characteristics to measure Business Process Quality.

Year 2012: Defining Process Performance Indicator by Using Templates and Patterns.

The article entitled: An examination of the literature relating to issues affecting how companies manage through measures, from year 2005 (https://dspace.lib.cranfield.ac.uk/handle/1826/3035) was selected for analysis. The article was chosen for further analysis based on the analysis of the references of some articles.

After carrying out automatic and manual searches and searches on the references of articles, the process of selecting studies began, which will be described in the next sub-section.

2.2 Selection of the Primary Studies

The step of selecting primary studies is about assessing the relevance of the articles to the research questions. The inclusion criteria used to select articles are as follows:

IC1 – Studies published from 1 January 2004;
IC2 – Studies that describe metrics, measurements and performance indicators that make it possible to evaluate the results of business processes;
IC3 – Studies that present rules and guidelines or for constructing and/or using measurements for evaluating business processes.

The following criteria were used to exclude articles:

EC1 – Studies published before 1 January 2004;
EC2 – Tutorials or seminars or abstracts;
CE3 – Documents that have not been written in English will be deleted;
CE4 – Repeated studies (separate articles, but with the same or similar content) or of little relevance;
CE 5 - Documents that do not have the full text of the paper available on the internet and whose author(s) also cannot be contacted will be deleted;
CE6 – Documents that clearly deal with other matters not relevant to the purpose of this systematic review will be deleted;
CE7 – Articles that present measurements for evaluating models of business processes;
CE8 – Studies that are simply limited to the presentation of measurements, without describing how to calculate them or do not provide a guide on how to use them.

Table 2 shows, respectively, the Digital Library (Digital Library), the filters used (Filters), the number of returned items in each library (Results), the number of articles whose titles were analyzed (Checked) and the number of articles selected for review after a second reading (Selected).

Table 2. Results of the searches.

Digital library	Filters	Results	Checked	Selected
IEEE	From 1 January 2004	342	342	10
ACM	From 1 January 2004	827	827	1
SpringerLink	From 1 January 2004	976	976	4
Science Direct	From 1 January 2004	512	512	1
Emerald	From 1 January 2004	720	720	9
TOTAL	From 1 January 2004	3,377	3,377	25

Having made an initial selection of the articles, a second filter was used. This consisted of making a a more thorough and detailed search, which meant reading, in full, the text of the articles initially selected in the automatic and manaua searches. In this phase, the help of 2 (two) other researchers to read the articles was requested. The decision whether or not include articles in the final selection had to be discussed by the 3 researchers. Table 3 lists all the articles used for this review which contributed to achieving the results. The first column shows the the IDs of the articles selected and the second gives the titles of the articles.

Tables 2 and 3 present the approaches investigated related to the evaluation criteria described above.

Table 3. Final result of selecting the studies.

ID	Title of article
EP6	Optimizing process performance visibility through additional descriptive features in performance measurement
EP7	Organizational performance measures for business process management: a performance measurement guideline
EP9	Research on key performance indicator (kpi) of business process
EP11	The research of metrics repository for business process metrics
EP13	Two cases on how to improve the visibility of business process performance
EP15	Performance measurement in business process outsourcing decisions: Insights from four case studies
EP22	Quality evaluation framework (QEF): modelling and evaluating quality of business processes

2.3 Critical Analyses of the Approaches

The different approaches discussed above show variations in the methodology, specifications and even as to how business processes are evaluated. In order to assess these approaches, a set of criteria was defined which addresses both the theory, based on the selected references, and some aspects of usability. The criteria used were:

- Methodology: Ad-hoc (Ah) OR Systematic (Sy);
- Types of measures: Quantitative (Qt) AND/OR Qualitative (Ql);
- Context of Applications: Specific (S) OR Generic (G);
- Processes supported by systems: Yes OR No;
- Efficiency: Yes OR No;
- Effectiveness: Yes OR No;
- Empirical validation: Yes OR No.

The criterion of Methodology is related to the attention to the way in which an approach can be used to evaluate a business process. Approaches classified as systematic ones are those that describe a set of rules, guidelines, processes or activities needed to use the measures presented. On the other hand, ad-hoc approaches focus only on describing performance measures, without, at first being interested in the way that such approaches can be put into practice.

The criterion called types of measures concerns the nature of the measures presented in the selected approaches. Quantitative measurements are those based on numerical performance indicators, while qualitative measures consist of textual descriptions and narratives about factors of success of the process and which often require interpretation.

The criterion called application is related to the context in which a particular approach can be used. Certain approaches present measures that are sufficiently generic so that they that can be applied in different contexts and business processes of very different natures. Approaches classified as specific are those used in a specific business process or those of a similar nature.

Some approaches have performance measures that are established from information generated by business process automation systems. In other words, the use of the measures presented is associated with making information available by means of systems. The criterion of processes supported by systems is related to the concern for analyzing if the performance evaluation of the process depends, in principle, on some support by means of systems such as a BPMS (Business Process Management Suite), for example.

The aim of using criteria of efficiency and effectiveness is to describe whether the approaches analyzed present measures to evaluate the productivity and performance (efficiency) of processes, as well as their ability to do what is needed, which is correct in order to reach a certain goal or outcome (effectiveness). Efficiency involves the way in which an activity or process is performed; effectiveness refers to whether this results in meeting customer's needs in all their restrictions. In [14], efficiency is a measure of the extent to which an organization's resources are used economically, and effectiveness refers to the extent to which the objectives are achieved.

The criterion of empirical validation is concerned with assessing the practical utility of the measures proposed in the different approaches. Empirical validation occurs by conducting experiments, case studies or research in a real context, and helps to determine the effectiveness of these measures.

In the seven approaches investigated, only two describe a systematic and procedural way, guidelines using the performance measures presented. The other studies are much more concerned with describing the performance measures of business processes than with how they can be used. The EP13 study presents some general guidelines for using performance measures, but does not describe them in detail.

All approaches presented quantitative performance measures, based on numerical performance indicators. On the other hand, only two articles, qualitative measures for evaluating the performance of processes, but these are about similar measurements, since both studies are by the same authors.

Regarding the context of the application, two approaches are linked to a specific domain, such as, for example, Article EP15 which presents measurements derived from specific outsourcing contracts, while five approaches have greater applicability.

Two articles describe performance measures dependent on information obtained because business processes had been automated, while five others do not mention the need to use systems in order to use the measures presented.

The seven approaches investigated describe performance measures to evaluate the efficiency of business processes. As to measures to evaluate the results of processes (effectiveness), only two articles do not describe them. Finally, three approaches were empirically validated by using case studies or applying them in a real context.

Tables 4 and 5 present the approaches investigated related to the evaluation criteria illustrated above.

Based on the criteria presented and discussed earlier in this paper, we considered that a systematic approach (which enables the metrics presented to be used consistently), which brings together quantitative and qualitative performance measures, generically (measures not linked to a specific context or domain), not dependent on using systems that provide an evaluation of efficiency (resources) and effectiveness

Table 4. Evaluation criteria (Methodology, Measures and Application).

Articles	Evaluation criteria					
	Methodology		Measures		Application	
	Ah	Sy	Qt	Ql	S	G
EP6	X		X	X		X
EP7		X	X			X
EP9	X		X		X	
EP11	X		X			X
EP13	X		X	X		X
EP15	X		X		X	
EP22		X	X			X

Table 5. Evaluation criteria (System support, Efficiency, Effectiveness and Empirical validation).

Studies	Systems	Efficiency	Effectiveness	Validated
EP6	No	Yes	Yes	No
EP7	No	Yes	Yes	No
EP9	Yes	Yes	Yes	No
EP11	Yes	Yes	No	No
EP13	No	Yes	Yes	Yes
EP15	No	Yes	Yes	Yes
EP22	No	Yes	No	Yes

(results) and which had been validated empirically could be considered ideal, or the approach that best meets how to evaluate the performance of business processes.

From the analysis of the selected approaches, there was a need to propose a systematic approach that would combine relevant and complementary aspects of the studies that had been previously analyzed.

3 Approach Proposed for Evaluating the Performance of Business Processes

Of the approaches investigated, the one that comprises the largest number of requirements, set out in the previous section, is Article EP13. However, it does describe in great detail how this approach can be used. Moreover, the approaches presented in articles EP7 and EP22 do give a detailed description of the the process for using performance measures.

It was arising from these considerations that our approach was defined. It brings together the model of performance measures described in EP13, but using it as a guide to using the measures presented, and the procedural description defined in Article EP7.

3.1 Process for Evaluating the Performance of Processes

The evaluation process defined in EP7 is illustrated in Fig. 1 below:

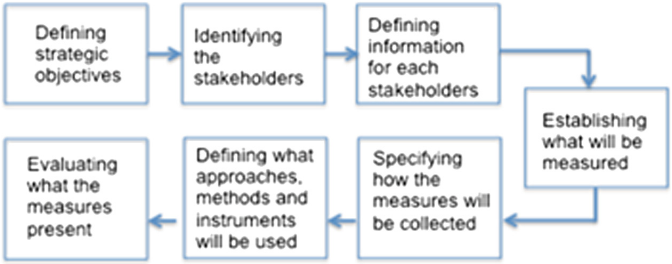

Fig. 1 Process for using performance measures.

The aim of the activities illustrated in the process flow is to lead to proposing a guideline for measuring the performance of business processes. In order to provide adequate guidance for assessing performance, on following this procedural description, a description will be given below of the steps and some artifacts or suggestions that may contribute to making such an assessment.

Defining the Strategic Objectives. According to Vuksicet et al. (2008), the performance measures of business processes should be aligned to the organizational goals and objectives. The first activity in the process of evaluating the performance of a process consists of defining the strategic objectives to be achieved by the results of the processes. These objectives are defined by the owners of the process in the organization, and can be obtained from the strategic planning documents.

Identifying the Stakeholders. After identifying the objectives of the process, the stakeholders in the conduct of the process must be identified, i.e., those responsible for the process, those taking part in the activities, clients, beneficiaries, sponsors and others. Process mapping artifacts or even the use of a RACI matrix can contribute to defining who the stakeholders in implementation are.

Defining the Information Required for Each Party. The information needs of each stakeholder can be very different, and require the performance measures to be adapted to their requirements.

The activity of mapping makes it possible to define the roles and responsibilities of each participant during the implementation of the process more clearly and objectively, thereby contributing to defining the information required for each party and assisting in defining the performance measures.

Establishing What will be Measured. In this stage what is sought is to identify what measures are useful for evaluating the performance of the process in relation to its goals, in achieving the strategic objectives. Performance reports or accounting documents can be used to identify performance measures of the process.

Specifying How the Measures will be Collected. When identifying the performance measures of a process, a process for collecting these measures needs to be identified. In the case of processes supported by systems, data collection is automated. In order to develop an action plan for identifying and collecting performance measures in non-automated processes, an adaptation of the5W2H management tool was proposed as an artifact for this step. The information is organized as follows:

- Description of the measure (What): Defines what this measure means.
- Rationale (Why): Defines the end, the purpose of using the measure.
- Instruments to obtain measures (From where): Specifies the instruments (media) from where the data of the measures can be collected.
- Frequency (When): Determines the frequency or time interval at which the information of the measures needs to be collected.
- Responsible (Who): Describes the sector or department responsible for providing information on a particular measure.
- Process for obtaining measures (How): Details what process will be undertaken to obtain the measures.
- Costs (How much): Describes the costs for obtaining the measure.

Defining What Approaches, Methods or Instruments will be Used. There are several approaches and frameworks that use different methods for measuring performance, described in the literature, as well as various tools that support these evaluation methods.

Based on the critical analysis of the approaches presented in Sect. 3, it was found that the model in EP13 has the largest number of criteria. Thus, it is recommended that this is used in step for evaluating the process. Subsection 3.2 presents how the measures in the EP13 model are systematized.

Assessing What the Measures Present. The performance measures of business processes should be related to (or the same as) measures existing in the organization that are used to monitor the success of the strategy.

3.2 Adapting the Evaluation Model

The performance measures defined in Article EP13 require a greater capacity of interpretation due to the way they are presented. The paper presents the measurements in the form of questions such as: Are the numerical parameters linked to quality, such as cycle time, probability of failure or average number of interruptions? This could hamper their use by individuals who evaluate their business processes.

Given this configuration, the measurement model was systematized in an attempt to make it orderly, coherent, clear and understandable by the parties that may use it. The model originally defined in EP13 was discussed together with two BPM experts until the model shown in Tables 6 and 7 was reached.

Also added to the model described above was an auxiliary table, which describes in detail each of the indicators presented. Such as, for example, "*Process Cycle time:*

Table 6. Model for systematizing the measurements (Efficiency).

	Efficiency	
NP (*)	***Indicators***	***Types of indicators***
	Key Performance Indicator (KPI)	- Time cycle of the process
		- Countable amounts of process outputs
		- Number of returned processes
		- Number of processes carried out
		- Downtime
		- Percentage of the budget used
DP (**)	Process Success Factors (PSF)	- Benefits, scope and steps of the process
		- Information systems, databases or interfaces used
		- Description of the success of the process
		- Documents and information produced and delivered

(*) Numeric Parameters; (**) Descriptive Parameters.

Table 7. Model for systematizing the measurements (Effectiveness).

	Effectiveness	
NP (*)	***Indicators***	***Types of indicators***
	Metrics of the Process (PMX)	- Probability of failure
		- Average number of interruptions
		- Time-cycle approval meetings
		- Number of approval meetings
		- Number of approvers/officers-in-charge
DP (**)	Description of the Components which comprise the Process (PO)	- Departments responsible, informed or affected
		- Role of approvers or officers-in-charge
		- Interfaces with other departments
		- Conditions and restrictions
		- Initiators, beneficiaries and customers of the process
		- Key people or information system for a step of the process or shared backup

(*) Numeric Parameters; (**) Descriptive Parameters.

corresponds to the time required for performing the process, i.e. the time between the start and end of the process".

The approach proposed for evaluating the performance of business processes dealt with in this study consists of the process for measuring performance, artifacts and the model of systematic measures described in this Section. In order to ensure the validation of the approach, it was applied in a real business process of an organization, and this is described in the next section.

4 Results and Discussion

The process that was selected to validate the proposed approach is called Pro-equipment, linked to the Pro-rectorate for Research and Postgraduate Subjects (PROPESQ) of a Brazilian Federal Institution of Higher Education (IFES, in Portuguese).

Pro-equipment procures equipment intended for shared use in the structure of scientific and technological research of post-graduate programs of the recommended IFESs and is funded by the Coordination Unit for Improving the Qualifications and Experience of Higher Education Personnel (CAPES). The selection of this process, to validate the proposed approach, occurred in a timely manner in view of the demand for having it mapped and optimized, requested through PROPESQ together with the IFES Office for Processes to which the author of this research is linked.

4.1 Defining the Objectives of the Process

The first interview was attended by two representatives of the Research Board (DPQ in Portuguese) and three process analysts of the Processes Unit of the institution were also present. The two participants are directly responsible for the performance of the process.

Initially a description of the following points was requested:

P1 – "What is the objective of mapping and optimizing the process?"

According to Respondent 2, the main problem of this process is the lack of transparency of the information on the results, when he states that:

E2 – "We have partial and not effective control, for example, we know the input volume of resources, we know how much CAPES makes available, and we even know how much of the resources were earmarked because the Accounts Department of PROPESQ send us this information on a spreadsheet, if we request it. From there on, we have no feedback, we do not know if the teacher received it."

P2 – "What is the Objective of the Pro-Equipment Process?"

E1 – "To stimulate scientific production by acquiring equipment intended for shared use in postgraduate laboratories of the university."

4.2 Identifying Stakeholders in the Process

P5 – "Who are the stakeholders in the process?" Specifically those in charge (the owners of the process), participants of the activities (sectors), clients, beneficiaries and sponsors.

CAPES acts as the sponsor, the clients of this process were defined as the teachers and/or research groups. Finally, the participants in the activities when the given process was being carried out were: the Director of Research (DPQ/PROPESQ), the

Accounting Board (DC/PROPESQ), the Agreements Sector (PROPLAN), the Legal Department (Office of the Rector), the Rector, The National Purchases Sector (PROGEST), the Importations Sector (PROGEST) and the Publishing Sector (PROGEST).

4.3 Defining the Information Necessary for Each Party

At this stage we were invited to the meetings, prior to which the stakeholders described above, excluding the sponsor of the process (CAPES). This was justified by the fact that CAPES is configured as an entity external to the University context in which one does not have dominion over the rules and procedures used.

Four meetings were held with the following representatives: 02 (DPQ/PROPESQ), 01 (DC/PROPESQ), 01 (Agreements Sector/PROPLAN) and 01 (PROGEST). At this stage the process was modelled collaboratively using the BizAgi Process Modeler.

The process starts when CAPES releases the official notice. Then, the Board of Research (DPQ) sets an internal schedule of activities before submitting a single proposal.

Generally, the amount of resources requested by the projects is greater than the amount initially made available by CAPES. Therefore, the DPQ convenes a committee to assess and recommend what projects should be submitted. The end result of the projects selected internally is announced and submitted to CAPES.

These projects will be further evaluated by CAPES, who may refuse some requests. Projects approved at this stage are announced by DPQ. After this step, the term of decentralization of credit is sent to PROPLAN, the process is reviewed, and the document is sent to the Legal Department, which will analyze items such as dates, terms, rubrics, data, etc. The Legal Department must give its assent in writing to ensure the process continues. If so, the process is sent for the signature of the Rector of the institution. The process for requesting ear-marking is then returned to PROPLAN which monitors that CAPES has released the funds and PROPLAN notifies the Accounting Sector of PROPESQ.

The Accounting Sector of PROPESQ then starts the activity of mounting the ear-marking process for each piece of equipment. At this moment, a request is made to the coordinators of the subprojects for a series of documents. Subsequently, the accounting entry for the committed funds is made via PROGEST, and there follows a new phase of analysis by the Legal Department. After being approved by the Legal Department, the process goes to the Publishing Sector in PROGEST, and the flow of the process proceeds to the National Purchases or Importations Sector so that the purchase can be made.

4.4 Defining What will be Evaluated in the Process

At this stage there were two interviews with those responsible (DPQ) for this process. Initially the interviewee was asked whether there is an accounting report on the performance of the process.

According to E1 - “CAPES demands an indication of how the previous buying process was conducted. We describe, when asked, only what was actually ear-marked”.

For interviewee E2, this kind of report is based on measures that do not reflect, in a satisfactory manner, that achieving the objectives of the process was verified.

E2 - “Ear-marking is only the first stage of the expenditure budget, where the funds were granted, but there is no guarantee the equipment will be purchased.”

When asked: “What are the important performance measures for making a satisfactory assessment of the results of this process?”

E1 - “In my opinion we should have information about whether or not the equipment was purchased, was actually installed in the laboratory and which research groups are using it”.

Still on the measures that should be used to evaluate the process, interviewee E2 stated that:

E2 - “Besides the amount ear-marked and bits and pieces to be paid, which are data that we can get easily, we need to know the quantity and description of the pieces of equipment bought, how many and which ones are awaiting delivery and installation, and what the institutional outcomes are (number of dissertations, theses, descriptive reports on patents generated or information on the rendering of research services to other institutions) that have been achieved”.

4.5 Specifying How the Measures will be Collected

At this stage a meeting was held with the same representatives who took part in the mapping of the process meetings. Initially, a projection of the mapping process was presented, followed by a set of measures that were extracted. The measures presented to the participants were:

- Total amount ear-marked;
- Amount spent the ear-marked resources;
- Amount in smaller bills to be paid;
- List of equipment purchased;
- List of delivered equipment;
- List of installed equipment and;
- Academic indicators.

For each of the measures proposed the artifact for specifying the collection process defined in the approach was discussed jointly by the participants and filled in. Table 8 shows an example.

4.6 Applying the Performance Assessment Model

At this stage, four meetings were held with the members already described: (02) representatives of DPQ/PROPESQ, (01) representative of the Accounting Sector/ PROPESQ, (01) representative of the National Purchases Sector and the Importations Sector of PROGEST. Meetings were held separately. The procedure for conducting this step may be described in three phases:

Table 8. Collection process for academic indicators.

Academic Indicators	
Description of measure	Corresponds to descriptive reports on data from the scientific production supported by the equipment purchased. Reports may contain descriptive information about numbers of dissertations, theses, patents obtained as a result of using the equipment, as well as descriptions of services rendered to other institutions
Justification	One of the strategic objectives of PROPESQ is to encourage scientific production in the university. Products purchased by Pro-equipment are intended to contribute to achieve this goal. Academic indicators enable the results of this process to be made more visible in relation to achieving the strategic objectives
Instruments to obtain measures	Scientific production reports from research laboratories that have equipment coming from Pro-equipment funds
Periodicity	*Ad hoc*
Officer-in-charge	Coordinators of postgraduate laboratories
Process for obtaining measures	Meetings or requests via email
Costs	Time

- Presentation of the Model – in this stage the presentation was made, using a print document format, of the model of measures defined to the interviewee.
- Recording of the participants' responses – the interviewees' answers were recorded in the document.
- Summary of the responses – in this stage, the responses were combined in a single Table, thereby eliminating redundancies and inconsistencies in a single frame.

Table 9 presents some of the indicators obtained in this step:

Table 9. Model for sytematizing the measures (Efficiency).

	Efficiency	
NP (*)	***Indicators***	***Types of indicators***
	Key Performance Indicator (KPI)	- Amount raised from CAPES - Amount of funds spent; ear-marked (Percentage used of the budget)
DP (**)	Process Success Factors (PSF)	- Steps: Official notice published, term of decentralization of credit undertaken, Review and analysis of the process, Release of Credit Authorized, Constructing ear-marked funds, review and analysis of the funds ear-marked followed by legal approval, Publication in the Official Gazette of the Union, Purchase and control of acquisitions made
		- Information systems, databases or interfaces used: SICAPES (the CAPES Integrated System), net purchases (Purchases Portal of the Federal Government), SICAF (System for the Unified Registration of Suppliers)

(*) Numeric Parameters; (**) Descriptive Parameters.

4.7 What the Measures Present

The process for measuring performance provides a better understanding of their real needs, and thus make it possible that better decisions and actions can be made in the future. After analyzing the literature and the case study, it would be valid to suppose that seems to be important that the measurements of business processes should be related to the measurements the organization already has that are used to monitor the success of the strategy.

A set of measures for the process as a whole, and its "partitions" (steps, departments, phases, etc.) were set out in the previous section so as to reflect certain performance characteristics for each interested management level. These measures describe by means of generating information the real state of the configuration of the process, thus enabling aspects of performance to be evaluated such as complexity in operations, bottlenecks, redundant activities, excessive documentation and approvals. In addition, they make it possible to evaluate the results of the process as to achieving the objectives set, according to some of the clients of this process.

Regarding Pro-equipment, a check was made on the opportunity to insert some improvements into the process. The following optimization proposals were discussed, as a result:

1. Integrating the Superintendence of Works (SPO in Portuguese) into the early stages of the process, specifically into the (internal) evaluation step of the sub-projects to be submitted to CAPES. The function of the SPO is to carry out assessment in order to identify if the laboratory to which the equipment will go, has the infrastructure needed for its installation. In several situations it was reported that the purchase of equipment had been made but the research lab did not have an infrastructure appropriate for its installation. Currently, the process for guaranteeing resources for renovations, or purchase of equipment to ensure the effective installation of the equipment is only started from the moment that the equipment is delivered to the university.
2. The urgency to seek the guarantee of funds for the purchase of equipment by ear-marking funds encouraged the actors of the department responsible not to check the (full) requirements of documentation defined by legislation for the formation of these processes (ear-markings). The process very often did not follow its "normal" flow, where and if it was published in the Official Gazette of the Union without necessarily obtaining the approval of the Legal Department. This deviation in the flow led, in various situations, to delays in conducting the process due to non-observance of the applicable legislation.
3. Information on the results obtained for the research based on acquiring equipment was practically non-existent or difficult to monitor. As a proposed solution which was discussed between the actors of the process is the creation of the role of research lab coordinator. The proposal is awaiting the approval of the rector of the university. One of these responsibilities is to draw up laboratory performance reports on its coordination in terms of research supported by the equipment purchased.

Organizations, in general, need to constantly check whether the performance of their business processes are compatible with the objectives set. For this there are different approaches or models using different methods to evaluate the performance of business processes. It appears from the results obtained in the systematic review of the literature that there is no single or universal approach that is the most appropriate evaluation of these processes.

At first, the adoption of the model, which combined quantitative performance measures (that permit performance to be measured and managed) and qualitative measures (which allow grounding the critical analysis of the results), manages to gather important information about the performance of the process in relation to the strategic goals and objectives set. However, it is seen because of the evaluation of the performance measures surveyed that the process which includes defining performance measurements directly associated with the strategic objectives set by those taking part in the process seems to be much more effective than using a static model of performance measures of generic processes.

From the case study analysis, it was also concluded that it is important that managers or owners of the processes have the understanding that will obtain relevant information about what they actually decided to measure, according to their assessment needs. In the case study evaluation, a check was also made on the need, for the participants of the process, to define performance measurements that are simple to interpret, measure and obtain. In this context, it was found that using very complex performance measures or those that are difficult to measure would not be appropriate, since the cost of obtaining them could adversely affect how doing so is made operational.

The risk in evaluating the approach in the context of a specific case should be noted and, in this case, the owners of the process may have only a partial view of the organizational goals and strategies, and how this process can achieve such goals. Thus, this requires a broader analysis between interrelated processes around similar strategic objectives and should have the complementary view of other participants in these processes.

The selected measures have not yet been implemented in practice, before writing this article, mainly due to the difficulty of change in a very short time interval in the public service context. To some extent, this adversely affected evaluating the effectiveness of the approach. On the other hand, the subsequent collection of the results of the process using the measures selected, can be considered a future opportunity for expanding this research.

5 Conclusions

The aim of the BPM approach is to provide pertinent information on running business processes, thus contributing to being able to make improvements and so that processes can be managed, thereby making better decision making possible. In this context, it is essential to use metrics that enable the initiative to be monitored, thus helping to verify how well the processes meet the strategic objectives set.

However the performance measurement initiatives do not seem to be as effective as evaluating the results of the process as to achieving the strategic objectives. This, therefore, hinders the capacity to visualize what the effects of BPM initiatives in organizations are.

In this context, the main objective of this research was to identify which model for evaluating the performance of business processes best fits the evaluation of business processes, with a view to a greater alignment between the indicators of the process and the strategic objectives.

With regard to reaching the overall goal, a systematic review of the literature was conducted in order to identify the approaches that had performance measures to evaluate results of business processes (Obj. 1). After carrying out the review, we identified seven studies that present and describe the measures used to evaluate the results of business processes. Then, a comparative analysis of approaches was made using criteria that consider theory and usability, assembled from the papers assessed (Obj. 2). At this stage, what was recognized was the need to develop a procedural approach to evaluating business processes based on complementary aspects of the approaches analyzed.

After defining the approach, an empirical study was conducted in order to apply it (Obj. 3), and to obtain an evaluation based on the perception of the leaders of the process as to its usefulness (Obj. 4). The results indicate that, in general, the approach was positively evaluated as to its effectiveness in providing relevant information on the performance of business processes in relation to the objectives set. Nevertheless, complementary assessments need to be made after the defined performance measures have been effectively used.

Regarding the overall objective of the research, which consisted of verifying which approach best serves the evaluation of business processes, with a view to greater alignment between the process indicators and the strategic objectives, it may be noted that there is no single approach that adequately assesses the performance of business processes.

The approach to measurement used should consider the organizational context and several variables can be measured and evaluated with regard to business processes. However, it falls to managers to undertake the tasks of identifying, selecting and defining measures that are adequate for and aligned with the organization's objectives. The model used has a large number of indicators that can and should be adapted to different organizational contexts, from which the most important for use in practice should be selected. Finally, it is essential that an organization uses several indicators when evaluating its business processes, since the use of a single indicator can hardly represent the broad context needed to support effective decision making.

In order to complement the results found in this research, we propose the approach should be used in several different business processes, with a view to verifying its real results for the different characteristics of operations and dynamics of an organization such as: culture, size, area of activity, and so on.

References

1. ABPMP: Guia para o Gerenciamento de Processos de Negócio - Corpo Comum de Conhecimento - (BPM CBOK®). Versão 2.0 (2009)
2. Smith, G., Furt, S.: How (not) to fail at BPM. In: BPM.COM (2009). http://bpm.com/how-not-to-fail-at-bpm.html
3. Trkman, P.: The critical success factors of business process management. Int. J. Inf. Manag. **30**(2), 125–134 (2010)
4. Powell, S.G., Schwaninger, M., Trimble, C.: Measurement and control of business process. Syst. Dyn. **17**, 63–91 (2011)
5. Kueng, P., Krahn, A.: Building a process performance measurement system: some early experiences. J. Sci. Ind. Res. **58**(3–4), 149–159 (2001). National Institute of Science Communication and Information Resources, New Delhi
6. Škrinjar, R., Bosilj-vukšić, R., Indihar-štemberger, M.: The impact of business process orientation financial and non financial performance. Bus. Process Manag. J. **14**(5), 738–754 (2008)
7. Pidun, T., Felden, C.: Two cases on how to improve the visibility of business process performance. In: 2012 45th Hawaii International Conference on System Science (HICSS), pp. 4396–4405, 4–7 January (2012)
8. Leclair, C., Cullen, A., Keenan, J.: Use a metric framework to drive BPM excellence. In: Forrester (2012)
9. González, L., Rubio, F., González, R., Velthuis, M.: Measurement in business processes: a systematic review. Bus. Process Manag. J. **16**(1), 114–134 (2010)
10. Wohlin, C., et al.: Experimentation in Software Engineering – An Introduction, 1st edn. EUA: Kluwer Academic Publishers, Berlin (2000)
11. Sampaio, R.F., Mancini, M.: Systematic review studies: a guide for careful synthesis of the scientific evidence. Revista Brasileira de Fisioterapia **11**(1), 77–82 (2007)
12. Kitchenham, B.: Procedures for performing systematic reviews. Joint Technical report, TR/SE-0401 and NICTA 0400011T.1, Keele University (2004). http://www.idi.ntnu.no/emner/empse/papers/kitchenham_2004.pdf
13. BPM CBOK – Guia para o Gerenciamento de Processos de Negócio. Corpo Comum do Conhecimento – ABPMP BPM CBOK V3.0, Association of Business Process Management Professionals (2013)
14. Corrêa, H.L., Corrêa, C.A.: Administração de Produção e Operações, 2nd edn. Atlas, São Paulo (2006)